



General Information
How to Use This Manual



Performing a Structural Repair on a Tesla Vehicle
Personal Protection
Sectioning Guidelines
Removing a Component
Preparing for Installation
Installing the Replacement Component
Post-Repair Operations
Contacting Tesla Collision Repair
Recent Changes In This Manual



General Repair Guidelines
Body Structure Materials and Allowed Operations
Dimensional Specifications
Electrical Harness Repair Guidelines
Fixed Glass Installation Preparation
Plastic Component Repair Guidelines
Preparing a Total Loss Vehicle for Transport
SRS Inspection and Component Replacement
Steering Gear Replacement Guidelines
Structural Repair Guidelines



Approved Parts, Tools, and Supplies
Approved Chemicals
Approved Fasteners and Fastener Tools for Structural Repairs
Approved Frame Bench Systems and Electronic Measuring Systems
Approved Gas Metal Arc (GMA) Welders and Welding Wire
Approved Paint Systems for Tesla Warranty Repairs
Approved Squeeze-Type Resistance Spot (STRS) Welders
Microstop Countersink Kit
Repair Guidelines for Parts, Fasteners, and Chemicals
Toolbox Labor Times
Using Flow Form Rivets



Part Repairability Guidelines
Approved Wheel Repairs
Front Bumper Fascia Repair Guidelines
Front Frame Rail Inner Repair Guidelines
Windshield and Roof Glass Repair and Replacement Guidelines

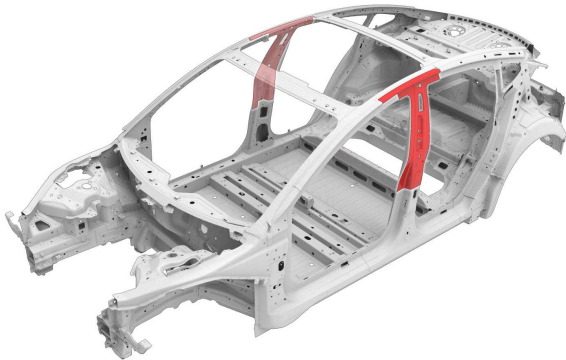


Reference Information
Vehicle and Wheel Paint Codes



Front Inner Panels
Cowl Lower
Cowl Top
Crossmember Sealing Platter (Front)

B-Pillar Outer (Upper)



Correction code: 101001382I

10100138302 **NOTE:** Unless explicitly stated in the procedure, the above correction code includes all Collision Repair and Service repair work required to perform this procedure, including the linked Collision Repair procedures and linked Service procedures. **Do not stack Collision Repair correction codes unless explicitly told to do so.** Depending on the damage to the vehicle, additional repairs may be required.
Parent topic: [Body Side Panels](#)

Repair Information

- Review all collision repair [general practices and safety documentation](#) and wear the appropriate [PPE](#) (Personal Protective Equipment) before beginning this procedure.
- [Properly mount the vehicle on a frame bench](#) when performing this procedure.

Parts List

Quantity Description

	Quantity	Description	
Dash Crossmember			
Dash Reinforcement Assembly			
Fender and Lamp Support Bracket			
Fender and Lamp Support Bracket Assembly			
Fender Support Bracket			
Front Frame Rail (Complete)			
Front Frame Rail Inner (Small Front Section)			
Front Frame Rail Inner Reinforcement			
Front Frame Rail Outer			
Front Shock Tower Cap			
Front Shock Tower Support (Front)			
Front Shock Tower Support (Rear)			
Front Subframe Inner Mount			
Front Subframe Mount (Rear)			
Front Subframe Mount Cap			
Frunk Bracket			
Longitudinal Upper			
Shotgun Inner	1	ASY - DOOR RING OUTER LH (Door Ring Outer Assembly)	
Shotgun Outer			
Shotgun Shear Wall			
Shotgun Tower			
Toeboard Outer			
Torque Box Cover Assembly			
Torque Box Rear Upper			
▼			
Body Side Panels			
A-Pillar Outer Reinforcement			
A-Pillar to Shotgun Reinforcement			
B-Pillar Inner Assembly (Complete)			
B-Pillar Outer (Complete)			
B-Pillar Outer (Lower)			
B-Pillar Outer (Upper)			
Body Side Outer			
Body Side Outer Front Cap			
Cantrail Inner Assembly	24	● Structural Bulb Rivet, 6.5 mm	
Cowl Side Assembly			
Door Ring Outer		●	
Fender Attachment Bracket (Lower)	17	● Countersunk Rivet, 4.8 mm Short	
Fender Attachment Bracket (Middle)			
Fender Attachment Bracket (Upper)			
Hinge Pillar			
Quarter Outer Assembly (Without Upper Trough)			
Quarter Outer Skin			
Roof Rack Brackets			
▼			
Center Under Body Panels			
Center Underbody			
Console Mount (Front)			
Console Mount (Rear)			
Front Floor Extension			
Front Seat Mount Reinforcements (Front Inboard and All Rear)			
Front Seat Mount Reinforcements (Front Outboard)			
Heelboard Crossmember (Front)			
Heelboard Crossmember (Rear)			
Rear Inner Sill Extension			



When ordering parts, refer to the [Parts Catalog](#) and enter the VIN of the vehicle being repaired to find the

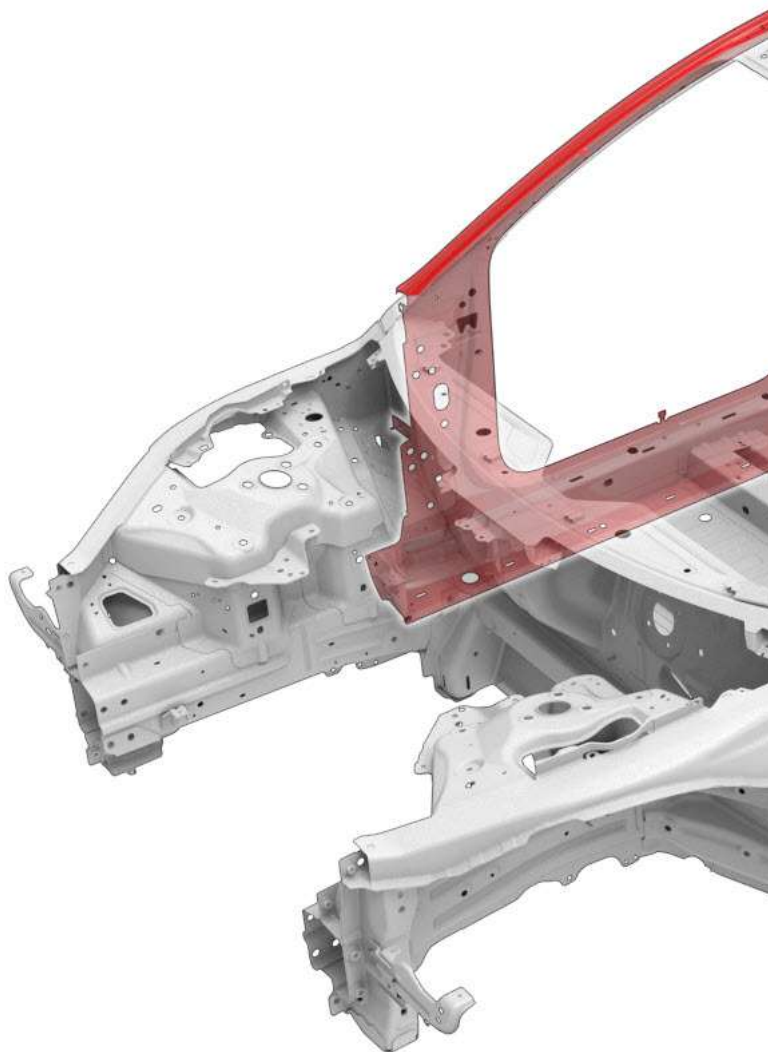
Repair Procedure

1. Remove the [Airbag – Curtain – RH \(RHD\) \(Remove and Replace\)](#).
2. Remove the [Retractor – Seatbelt – Driver \(Remove and Replace\)](#).
3. Remove the [Carpet – Rear \(Remove and Replace\)](#).
4. Remove the [Carpet – Front – LH \(Remove and Replace\)](#).

Remove the necessary portion of the [Body Side Outer](#) to expose the underlying component being replaced.

5. replaced.

Rear Jack Pad Bracket
 Sill Inner (Complete)
 Sill Outer (Complete)
 Sill Outer (Rear)
 Subframe to Sill Closeout
 ✓
 Rear Quarter Inner Panels
 Cantrail Inner Extension
 C-Pillar Reinforcement
 Divider Bar Reinforcement
 Parcel Shelf Crossmember Assembly (Without Center Trough)
 Parcel Shelf Extension (Front)
 Parcel Shelf Extension (Rear)
 Rear Center Trough
 Rear Damper Mount Upper
 Rear Damper Mount Upper Assembly with Stiffeners
 Rear Quarter Inner (Assembly)
 Rear Quarter Inner (Panel)
 Rear Wheel Arch (Complete)
 Rear Wheelhouse Extension
 Rear Wheelhouse Inner Complete Assembly
 Rear Wheelhouse Reinforcement
 Roof Rail Extension
 ✓
 Rear End Panels
 AMP Bracket
 Charge Port Controller Bracket
 Charge Port Housing Assembly
 Lamp Can Assembly
 Rear Aero Shield Attachment Bracket (Rear)
 Rear Aero Shield Harness Attachment Bracket
 Rear Body Panel
 Rear Frame Rail (Lower)
 Rear Frame Rail (Upper)
 Rear Frame Rail End Plate
 Rear Frame Rail Glove (Lower)
 Rear Frame Rail Glove (Upper)
 Rear Frame Rail Reinforcement (Lower)
 Rear Frame Rail to Lower Quarter Extension
 Rear Motor Cover Floor
 Rear Node (Complete)
 Rear Quarter Outer (Lower)
 Rear Spring and Subframe Mount
 Rear Subframe Crossmember (Lower)
 Rear Subframe Crossmember (Upper)
 Rear Subframe Mount
 Third Row Seat Crossmember (Lower)
 Third Row Seat Crossmember (Upper)
 Trunk Floor Assembly (Complete)
 Trunk Floor Assembly (Side Panel)
 Wheelhouse Outer Extension
 ✓
 Roof Panels
 Front Header Lower
 Front Header Reinforcement Connector
 Front Header Upper



6. Cut the new Door Ring Outer Assembly as indicated below to make the new component.

Rear Header Lower

Rear Header Reinforcement Connector

Rear Header Upper



7. Remove the original component.

 or  Factory Spot Welds





8. Prepare for installation.

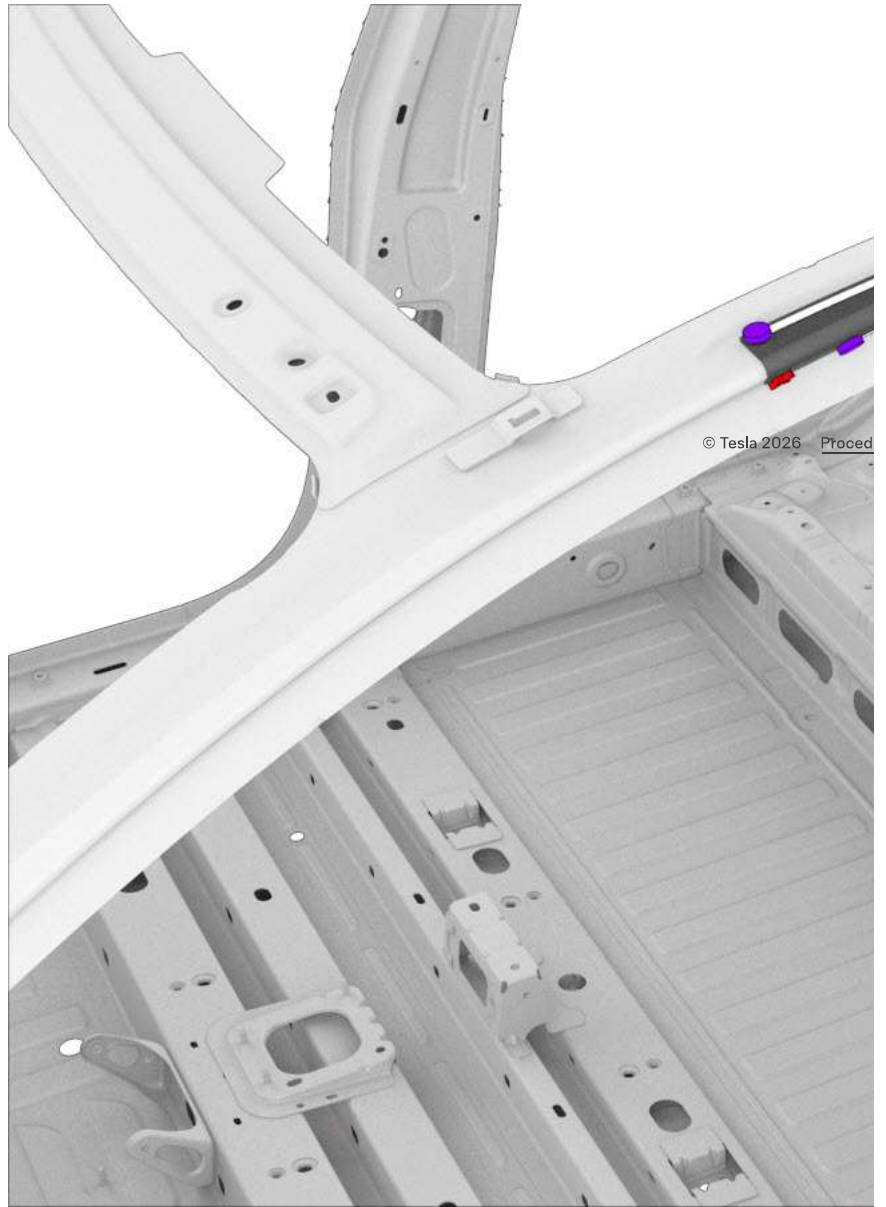
Note

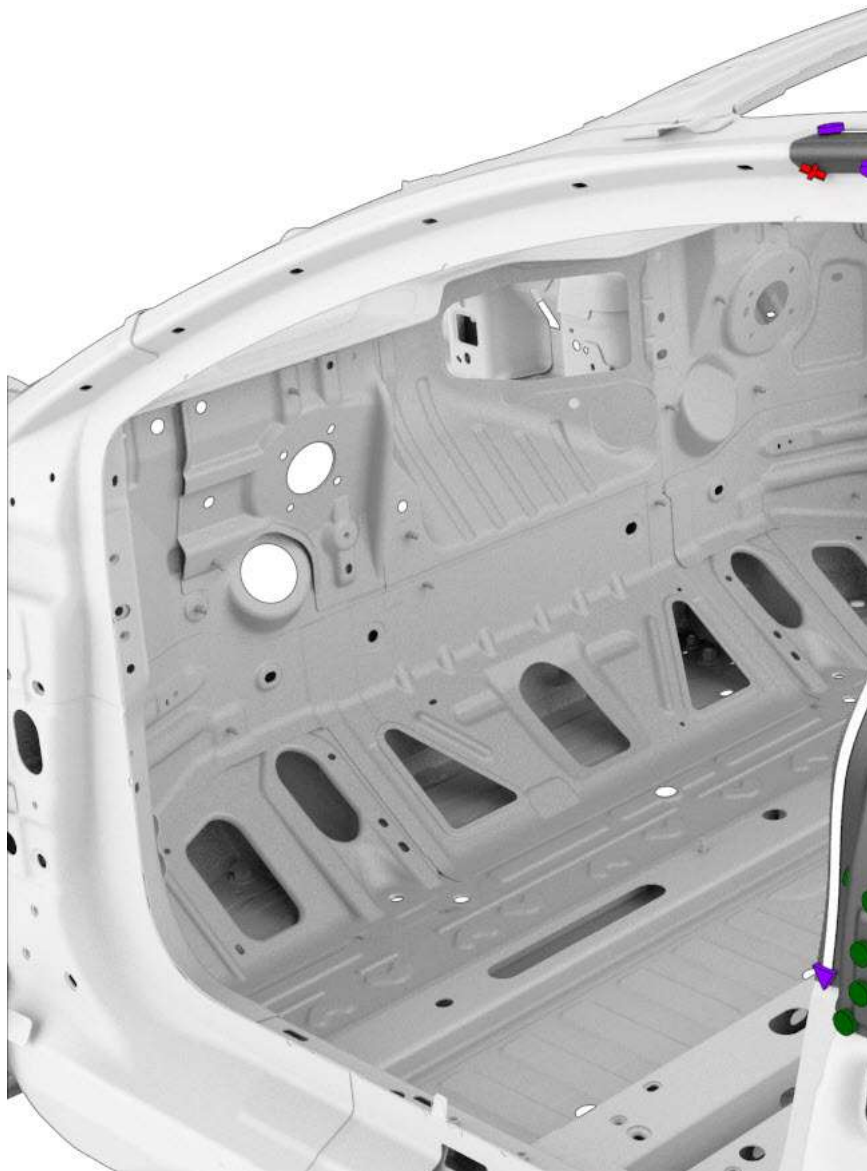
A red X indicates a location where a factory-installed fastener is not being replaced. Secure the location using structural adhesive only.

 or  Installation Spot Welds

 or  Structural Bulb Rivets, 6.5 mm

 or  Countersunk Rivets, 4.8 mm Short





9. Apply structural adhesive to the mating surfaces on the vehicle and the new component or component.
10. Install the new component or components.
11. Perform resistance spot welding.

 or  Installation Spot Welds

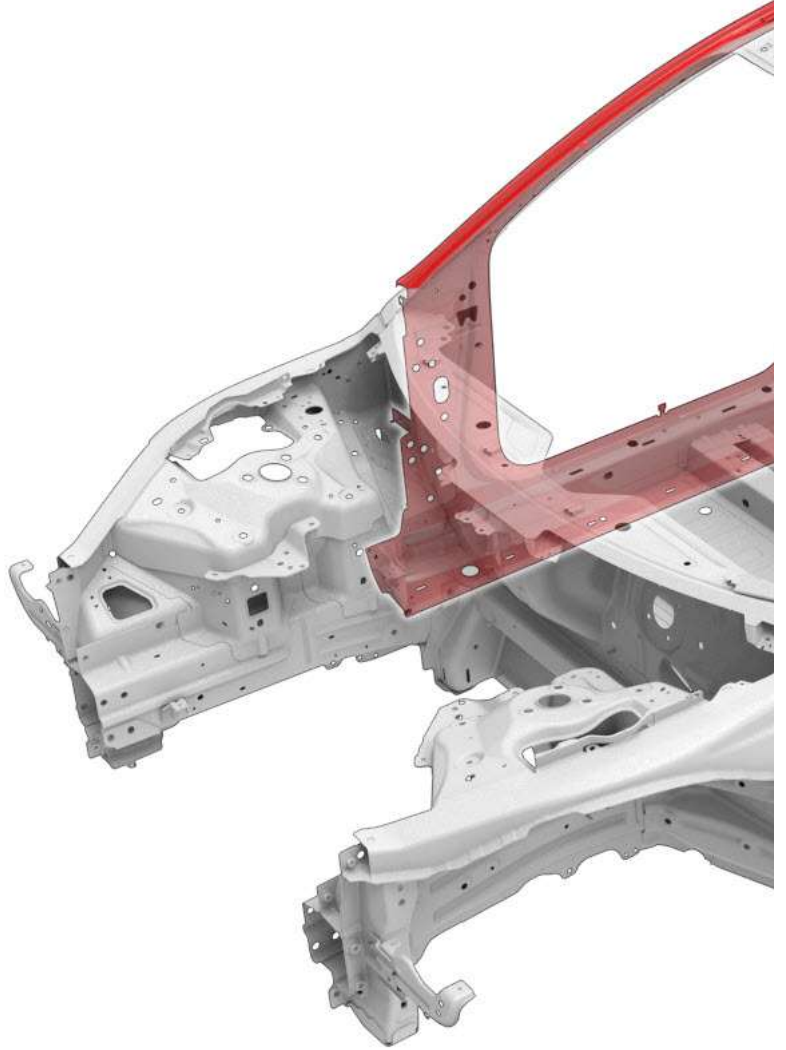
 Warning

Failure to follow all welding safety precautions, including the use of personal protective equipment, could result in serious injury or property damage. Only technicians who have completed Tesla's approved welding training are authorized to weld structural components on Tesla vehicles.

 CAUTION

Do not weld on a Tesla vehicle before performing the Vehicle Electrical Isolation Procedure (refer to the vehicle-specific Service Manual for more information on the Vehicle Electrical Isolation Procedure). Welding on a Tesla vehicle with an energized high or low voltage system might damage vehicle components.

12. Install the portion of the Body Side Outer that was removed to expose the component being replaced



13. Perform any necessary post-repair operations.
14. Install the Carpet - Front - LH (Remove and Replace).
15. Install the Carpet - Rear (Remove and Replace).
16. Install the Retractor - Seatbelt - Driver (Remove and Replace).
17. Install the Airbag - Curtain - RH (RHD) (Remove and Replace).